

# AUTOMATED AIR CARGO PACKING

AE4ASM527 Spatial AI for Industrial Automation (2025/26 Q4)  
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## PROBLEM STATEMENT

### Problem formulation:

Manual air cargo loading is slow, labor-intensive and prone to unstable stacking. An automated system is needed that can perceive boxes of varying sizes, plan stable arrangements and execute precise pick-and-place operations without human intervention.

### System function:

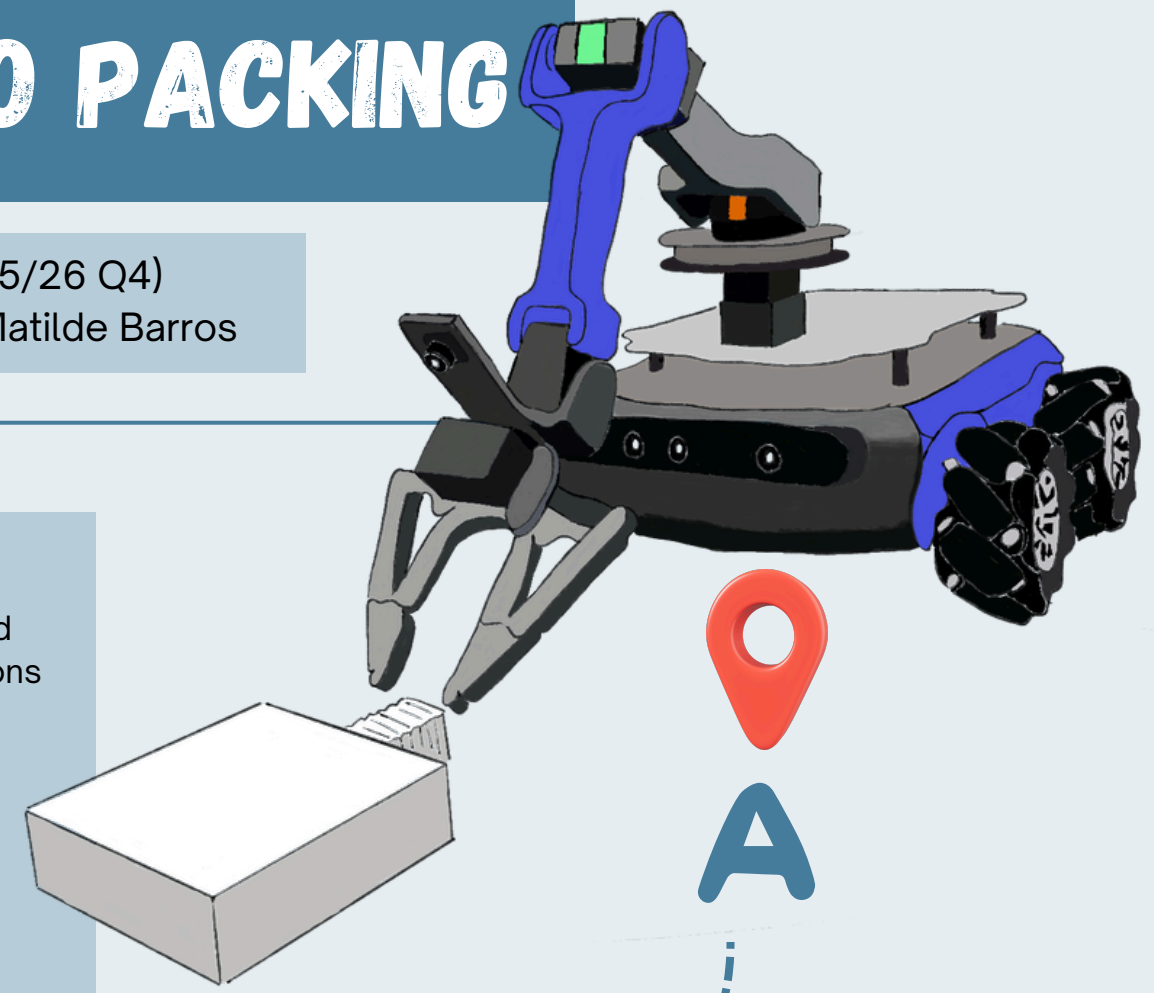
A robotic system that detects, localizes and stacks square cargo boxes from a pickup area (Station A) into an organized, space-efficient stack at a loading area (Station B).

### Constraints:

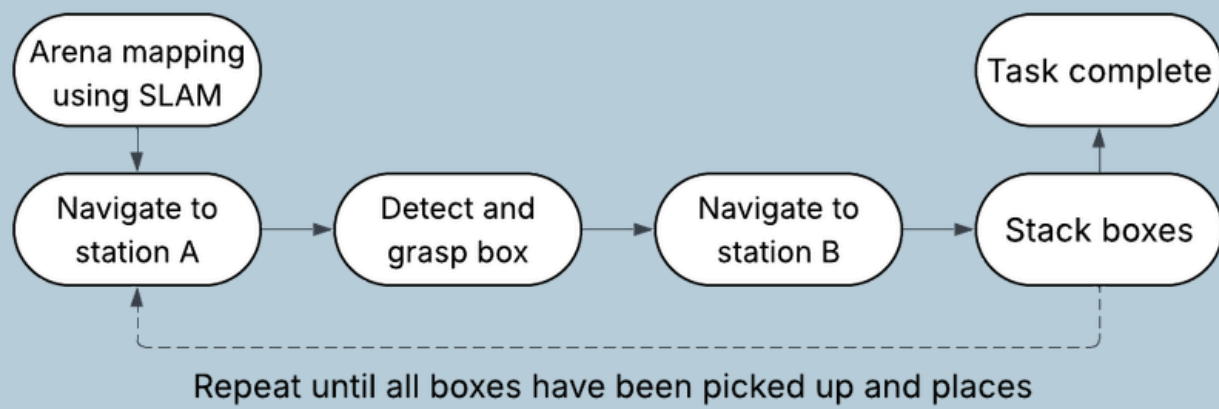
- No modifications to the boxes
- No knowledge of the arena layout during development
- Unknown starting position

### Key Challenges:

- Autonomous decision-making
- Box & gripper detection
- Real time mapping and navigation



## SYSTEMS OVERVIEW



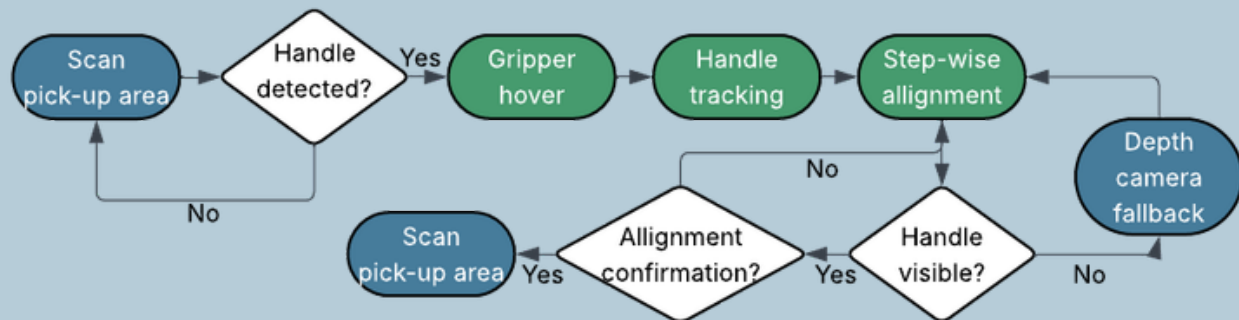
## PICKING UP STRATEGY

### Sensors used:

- **Depth camera:** wide field of view, used for workspace scanning and fallback localization
- **Gripper camera:** narrow, close-range, used for fine alignment

### Detection models:

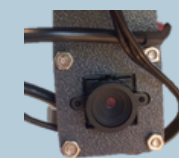
- Two YOLOv8n models trained on handle images captured from each camera



## SENSING & PERCEPTION



Depth/RGB Camera



Gripper Camera

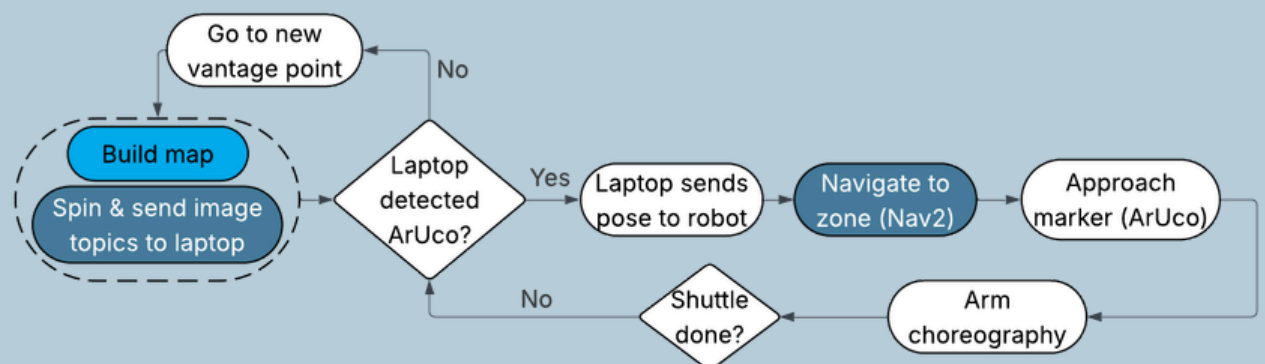


Lidar

## NAVIGATION

### Sensors used:

- **Lidar:** mapping with SLAM and obstacle avoidance during navigation
- **RGB camera:** streams compressed images to laptop for ArUco zone detection



## STACKING STRATEGY

1. **Positioning:** Move the gripper carrying the box directly over the target stack.
2. **Placement:** Slowly lower the box until it makes contact with the stack.
3. **Sliding Box:** Move the robot backward, sliding the box into its final position on the stack.
4. **Final Alignment:** Align the gripper with the other grippers using the ArUco markers.
5. **Release:** Open the gripper to release the box.

### Sensors used:

- **RGB Camera:** Orienting at position B and aligning grippers stack

## PACKAGES & RESOURCES

### Robot platform:

- MIRTE robot

### Perception:

- YOLOv8 (box handle detection)

### Navigation:

- Nav2
- SLAM Toolbox
- OpenCV (ArUco detection)

## CONCLUSION & FUTURE WORK

### Limitations:

- Faulty robot arm
- Slippery gripper
- Camera update rates

### Future work:

- Utilize computer vision to perfectly align the boxes after stacking
- Expand the problem to objects of varied shapes and sizes
- Optimize the stacking order and position of the objects

